Human Health and Disease

Dr. Savita Devi

Assistant Professor

Department of Biotechnology

IIT-Hyderabad

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భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Health:

- State of complete social, mental and physical well being



- Balanced diet, personal hygiene and regular exercise are very important to maintain good health
- Awareness about diseases and their effect on different bodily functions, vaccination (immunization) against infectious diseases, proper disposal of wastes, control of vectors and maintenance of hygiene in food and water resources are necessary for achieving good health

- When the functioning of one or more organs or systems of the body is adversely affected, characterized by appearance of various signs and symptoms, we say that we are not healthy, i.e., we have a **disease**

Classification of disease

Infectious

Easily transmits from one person to another
Examples: AIDS,

Non-Infectious

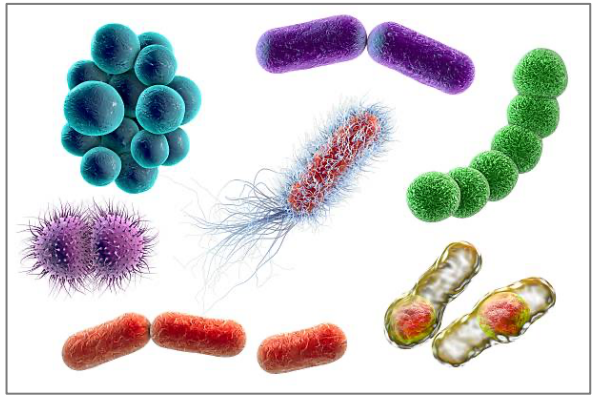
Do not transmits from one person to another
Example: cancer

Pathogens:

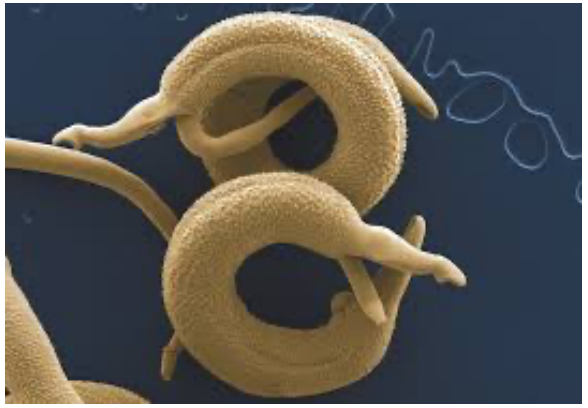
Diseases causing organisms are called pathogens.

Types of Pathogens:

Bacteria



Helminths



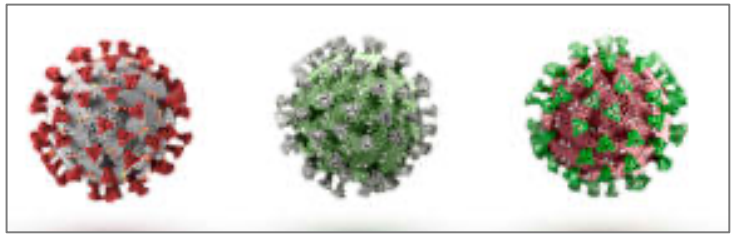
Fungi



Protozoans



Virus



Common infectious diseases

Bacterial diseases

Typhoid fever

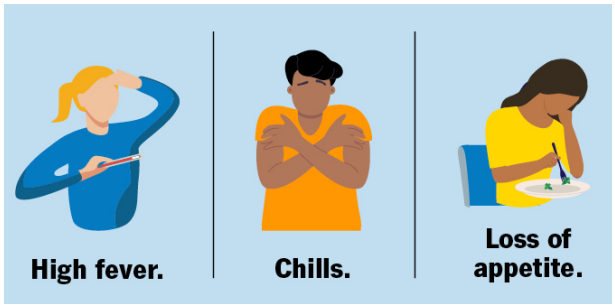
Salmonella typhi

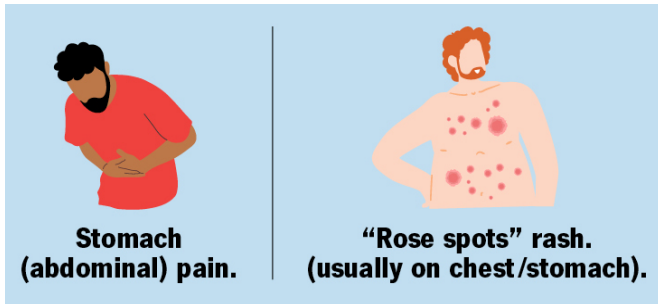


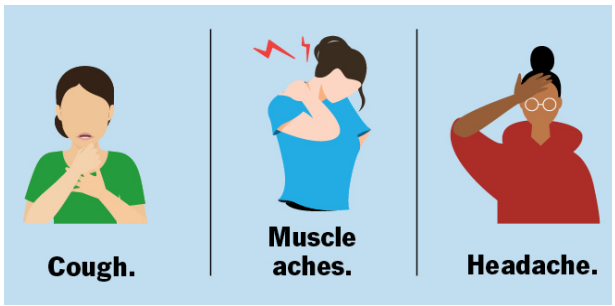
Pathogen: *Salmonella typhi*

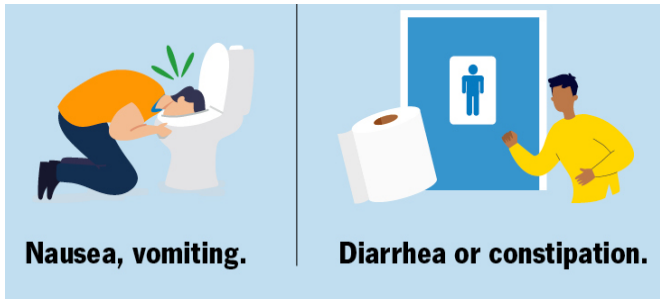
Mode of transmission: It enters small intestine through food and water and migrates to other organs through blood

Symptoms: High fever (39-40 ^{0}C), headache, weakness, stomach pain, constipation and loss of appetite

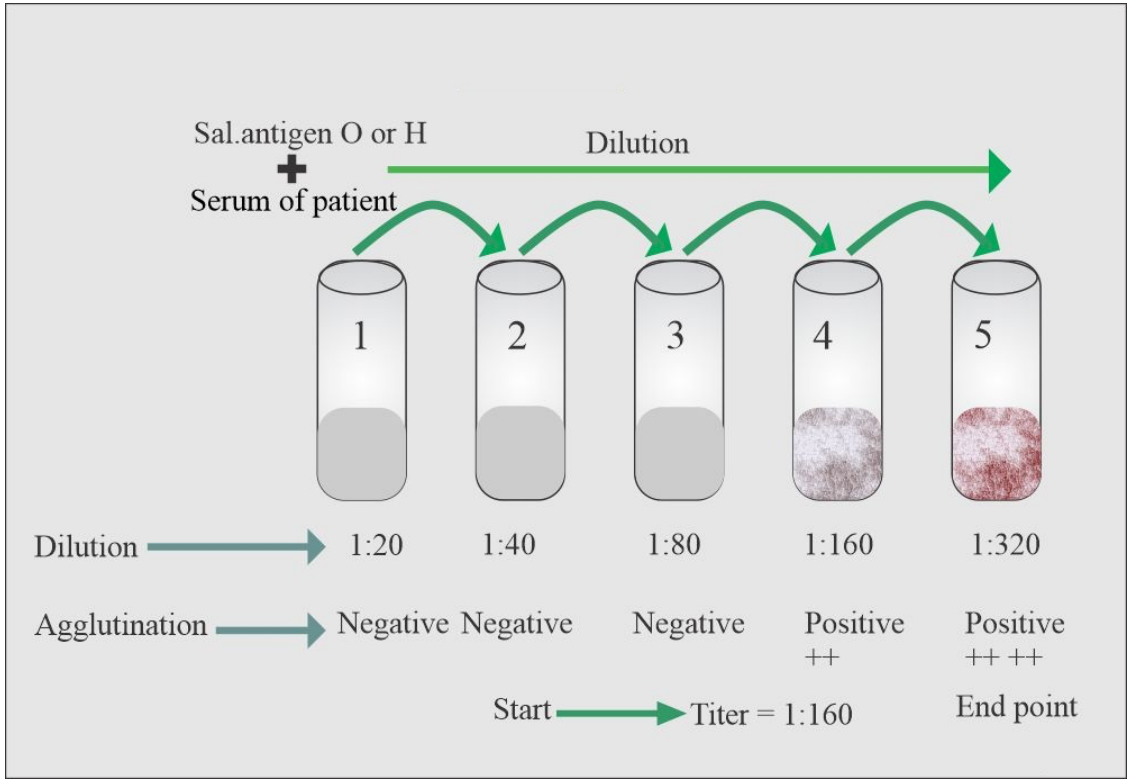


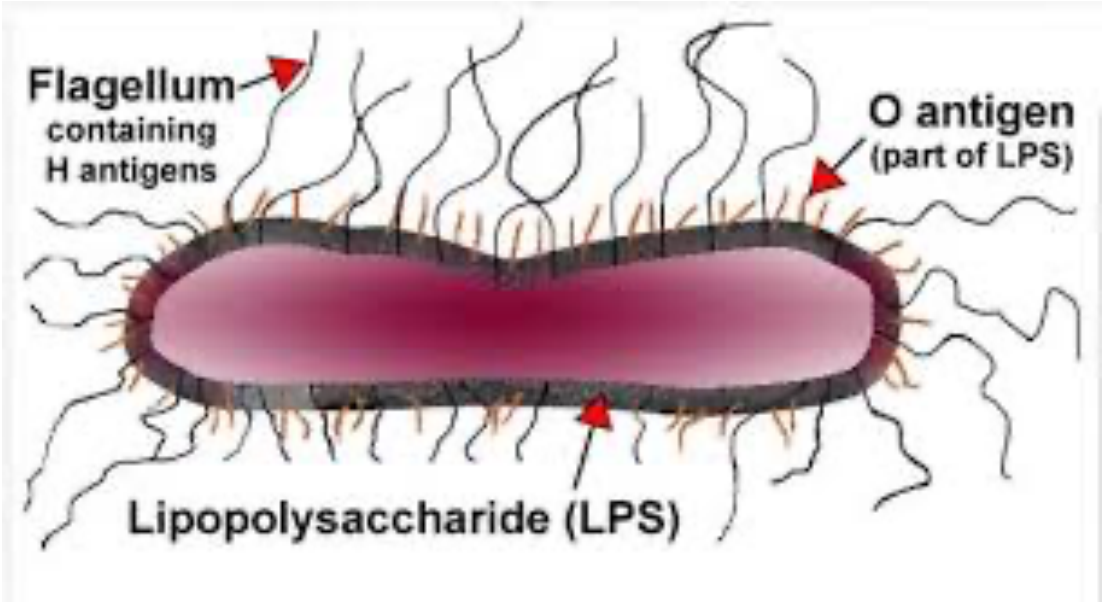






WIDAL test is used for the confirmation of Typhoid fever





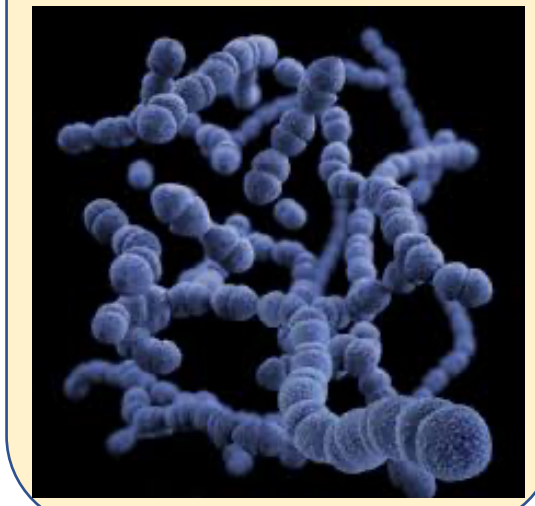
Agglutination: *Clumping of particles together. Aggregation of antigens and antibodies.*

Mary Mallon nicknamed Typhoid Mary was a professional cook. She was a Typhoid carrier who continued to spread typhoid for several years through the food she prepared.

Bacterial diseases

Pneumonia

Streptococcus pneumoniae



Pathogen: *Streptococcus pneumoniae, infect the lung alveoli (air filled sacs of lungs).*

Mode of transmission: A healthy person acquires the infection by inhaling the droplets/aerosols released by an infected person or even by sharing glasses and utensils with an infected person.

Symptoms: fever, chills, cough and headache, In severe cases, the lips and finger nails may turn gray to bluish in colour.

Other bacterial diseases:

Dysentery, plague, diphtheria

Viral diseases

Common Cold

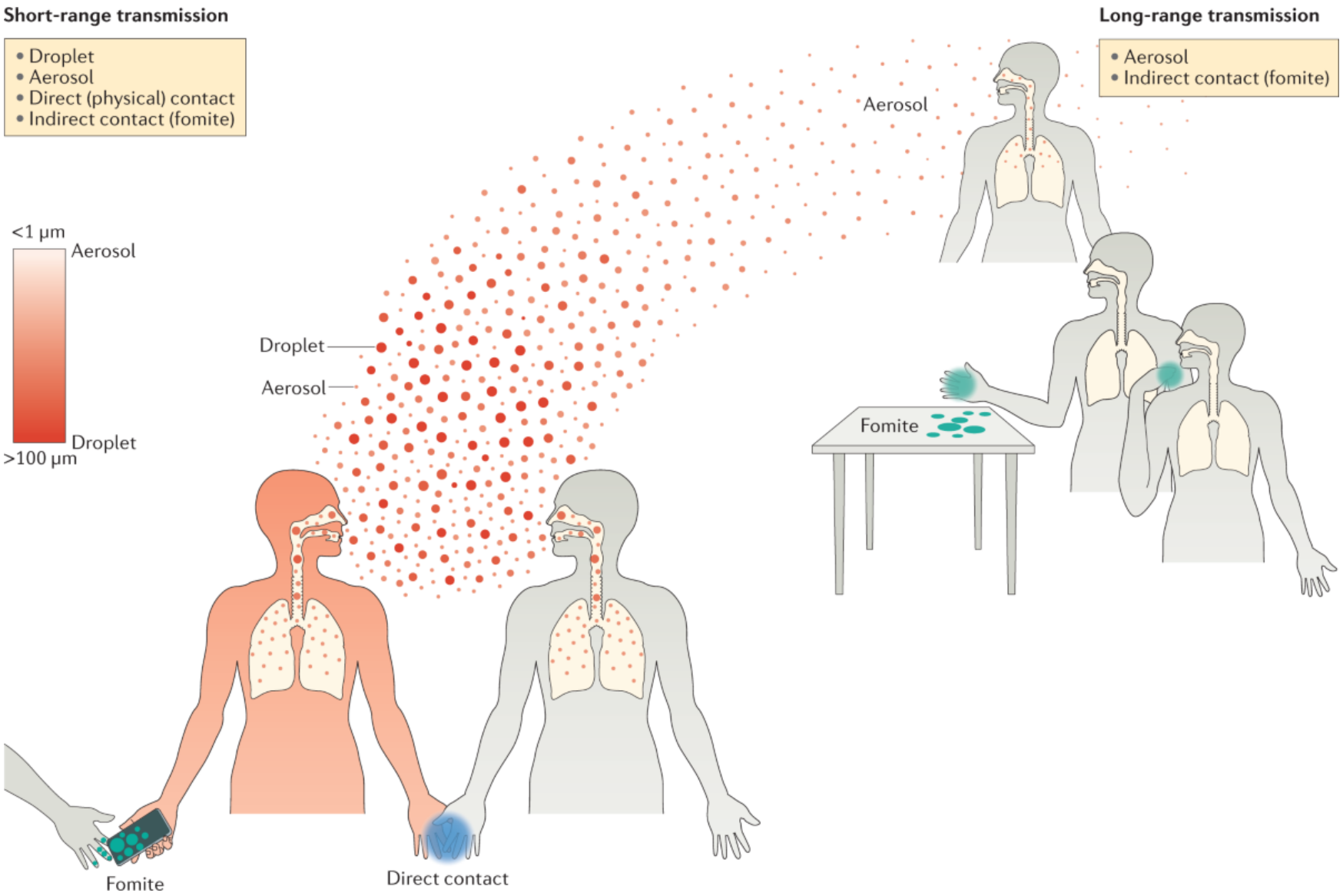
Rhino Viruses



Pathogen: Rhino Viruses, infect the nose and respiratory passage but not lungs.

Mode of transmission: Droplets resulting from cough or sneezes of an infected person are either inhaled directly or transmitted through contaminated objects such as pens, books, cups, doorknobs, computer keyboard or mouse, etc.,

Symptoms: nasal congestion and discharge, sore throat, hoarseness, cough, headache, tiredness, which usually last for 3-7 days.

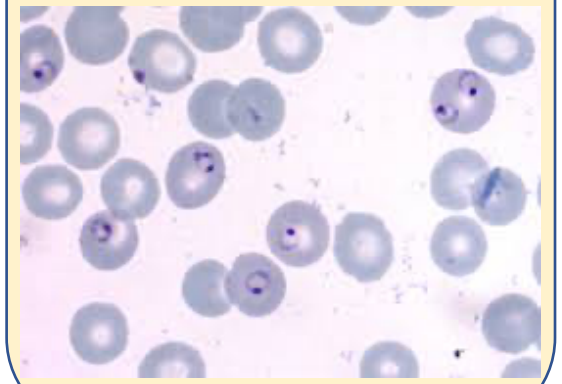


Fomite: Objects or materials which are likely to carry infection, such as clothes, utensils, furniture, pens etc.

Protozoan diseases

Malaria

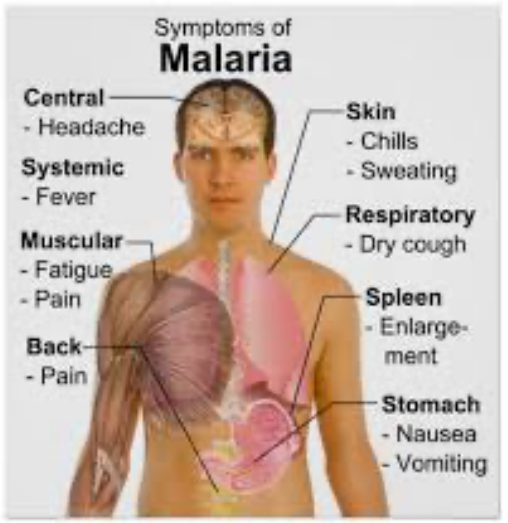
Plasmodium sp.

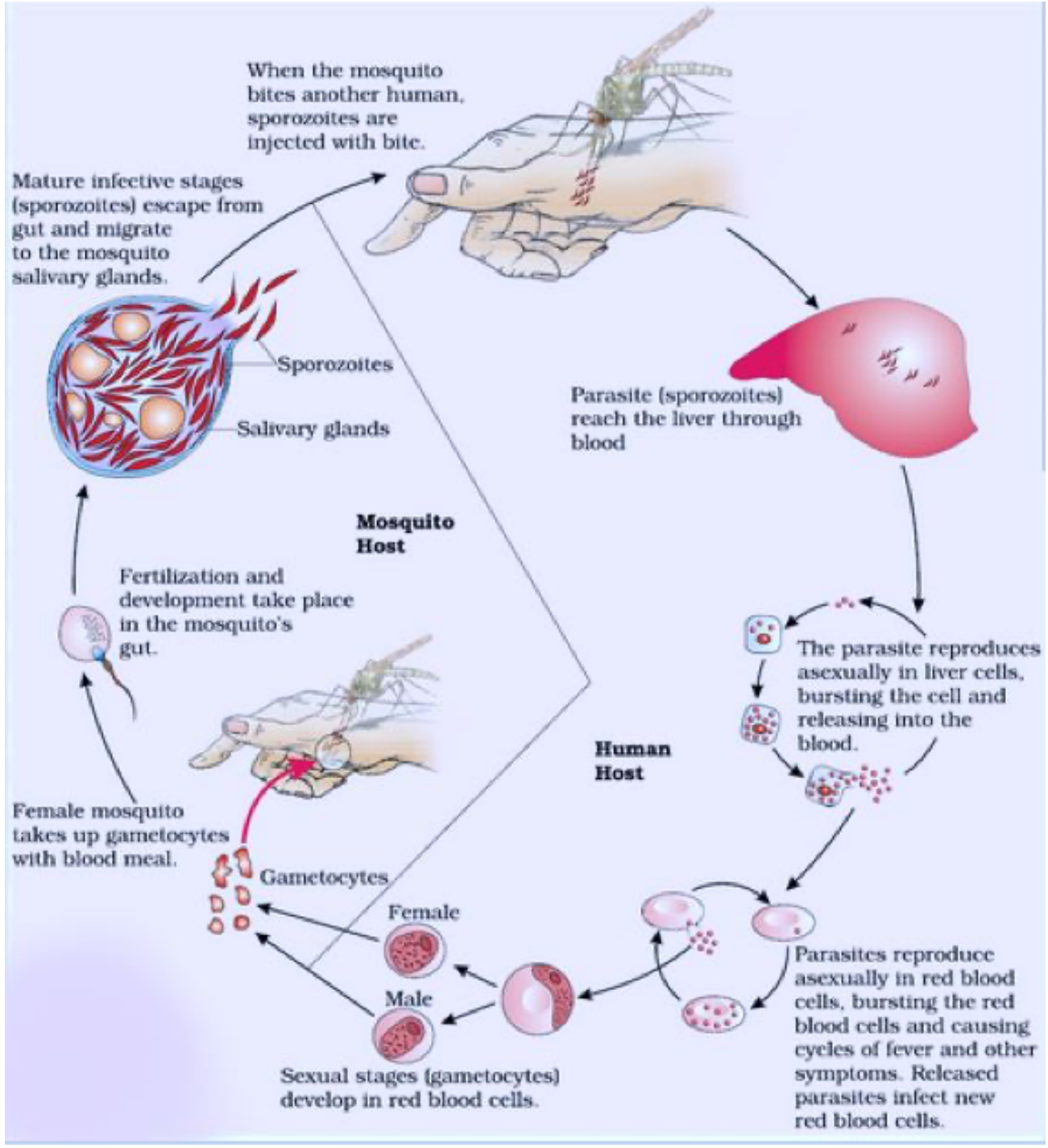


Pathogen: Plasmodium sp. (*P. vivax, P. malariae, P. falciparum*). *P. falciparum* causes most serious malaria

Mode of transmission: by biting of female Anopheles mosquito.

Haemozoin (toxin released by Plasmodium) causes chill and high fever recurring every 3-4 months.





Life cycle of Plasmodium

Sporozoites: Mature infective form of Malaria parasite

Protozoan diseases

Amoebiasis (Amoebic dysentery)



Pathogen: Entamoeba histolytica

Mode of transmission: Houseflies (mechanical carriers) transmit parasites from faeces to food and water.

Symptoms: Constipation, abdominal pain and cramps, stools with excess mucous and blood clots.



Helminth diseases

Ascariasis



Pathogen: *Ascaris* (Intestinal parasite)

Mode of transmission: soil, water, vegetables, fruits etc. contaminated with feaces containing eggs of parasites.

Symptoms: Internal bleeding, muscular pain, fever, anemia and blockage of intestinal passage.

Helminth diseases

Filariasis (Elephantiasis)



Pathogen: *Filarial worms or Wuchereria* (*W. bancrofti* and *W. malayi*)

Mode of transmission: bite of female Culex mosquito

Symptoms: Filarial worms live in lymphatic vessels (usually of lower limbs). It causes chronic inflammation of the organs in which they live for many years. Limbs and genital organs may be deformed.



Fungal diseases

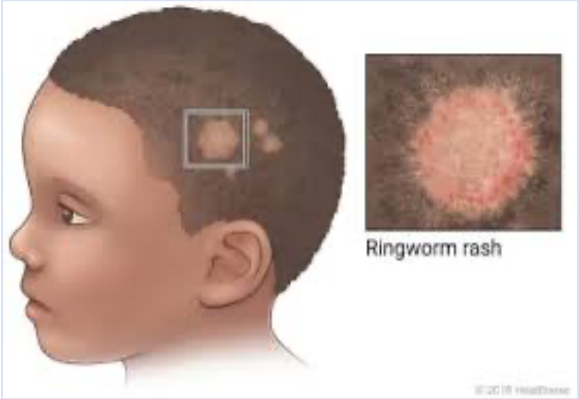
Ring worms



Pathogen: *Microsporum, Trichophyton and Epidermophyton. They are seen in groin, between toes etc.*

Mode of transmission: from soil or by using towels, cloths, comb etc. Heat and moisture help fungi to grow.

Symptoms: dry, scaly lesion on skin, nail, scalp etc. intense itching .



Prevention and control of diseases

- Personal Hygiene
- Keep the body clean.
- Use clean drinking water, food etc.





Public Hygiene

- Proper disposal of wastes and excreta
- Periodic cleaning and disinfection of water reservoirs, poolsm cesspools and tanks
- Avoid contact with infected persons or their belongings (to control air brone diseases)
- Standard practices of hygiene in public catering.
- Control and eliminate vectors (mosquitoes)

- Vaccines and immunization helped to control diseases like smallpox, polio, diphtheria, pneumonia and tetanus.

- Drugs like antibiotic helped to treat infectious diseases.







HUMAN IMMUNE SYSTEM



Immune system:

It is the system that gives immunity to the body by recognizing, responding and remembering foreign antigens.

Immunity:

Immunity: Derived from word *immunis* 'exempt': A state of protection from infectious disease.

Early vaccination studies led to the immunology: Edward Jenner (Small Pox)



- Immunity plays a role in allergic reaction, auto-immune disease and organ transplantation.
- It includes lymphoid organs, tissues, cells and antibodies



